LAB Assignment 4
Object-Oriented Programming

Name: Mahak Goyal
Roll No.: 20226010066
Course : MCA (2026-2028)
Group : 1MCA3

Q1. Create a C++ program to demonstrate Single Inheritance.

Create a base class Person with the following data members:
Name
Age
Create a derived class Student having:
Roll Number
Course
Implement member functions to accept and display the complete details of a student using inheritance.

```
#include <iostream>
using namespace std;

class Person
{
protected:
    string name;
    int age;
public:
    void acceptPerson()
    {
        cout << "Enter full name: ";
        getline(cin, name);
        cout << "Enter age: ";
        cin >> age;
    }

    void displayPerson()
    {
        cout << "Name: " << name << endl;
        cout << "Age: " << age << endl;
    }
};

class Student : public Person
{
private:
    int rollNo;
    string course;

public:
    void acceptStudent()
    {
        acceptPerson();
```

```
        cout << "Enter roll no: ";
        cin >> rollNo;
        cout << "Enter course: ";
        cin >> course;
    }

    void displayStudent()
    {
        displayPerson();
        cout << "Roll No: " << rollNo << endl;
        cout << "Course: " << course << endl;
    }
};

int main()
{
    Student s;
    s.acceptStudent();
    cout << "\nStudent Details:" << endl;
    s.displayStudent();
    return 0;
}
```

```
Enter full name: Disha
Enter age: 21
Enter roll no: 101
Enter course: MCA

Student Details:
Name: Disha
Age: 21
Roll No: 101
Course: MCA

--------------------------------
Process exited after 16.1 seconds with return value 0
Press any key to continue . . .
```

Q2. Create a C++ program to demonstrate Multilevel Inheritance.

Create three classes:
Person → stores Name
Employee → stores Employee ID
Manager → stores Department
Derive Employee from Person and Manager from Employee. Display all the details of the manager.

```
#include <iostream>
using namespace std;
class Person
{
protected:
```

```
    string name;

public:
    void acceptPerson()
    {
        cout << "Enter name: ";
        getline(cin, name);
    }
};

class Employee : public Person
{
protected:
    int employeeID;
public:
    void acceptEmployee()
    {
        acceptPerson();
        cout << "Enter employee ID: ";
        cin >> employeeID;
    }
};

class Manager : public Employee
{
private:
    string department;
public:
    void acceptManager()
    {
        acceptEmployee();
        cin.ignore();
        cout << "Enter department: ";
        getline(cin, department);
    }

    void display()
    {
        cout << "\nManager Details:" << endl;
        cout << "Name: " << name << endl;
        cout << "Employee ID: " << employeeID << endl;
        cout << "Department: " << department << endl;
    }
};

int main()
{
    Manager m;
    m.acceptManager();
    m.display();
    return 0;
}
```

```
Enter name: Disha
Enter employee ID: 1004
Enter department: Computer Science

Manager Details:
Name: Disha
Employee ID: 1004
Department: Computer Science

--------------------------------
Process exited after 46.95 seconds with return value 0
Press any key to continue . . .
```

Q3. Create a C++ program to demonstrate Multiple Inheritance.
Create two base classes:
Academic → stores Marks
Sports → stores Sports Score
Create a derived class Result that inherits from both classes and calculates the total score and displays the result.

```
#include <iostream>
using namespace std;

class Academic
{
protected:
    int marks;

public:
    void acceptAcademic()
    {
        cout << "Enter academic marks: ";
        cin >> marks;
    }
};

class Sports
{
protected:
    int sportsScore;

public:
    void acceptSports()
    {
        cout << "Enter sports score: ";
        cin >> sportsScore;
    }
};
```

```
class Result : public Academic, public Sports
{
public:
    void displayResult()
    {
        cout << "\nAcademic Marks: " << marks << endl;
        cout << "Sports Score: " << sportsScore << endl;
        cout << "Total: " << marks + sportsScore << endl;
    }
};

int main()
{
    Result r;

    r.acceptAcademic();
    r.acceptSports();

    r.displayResult();

    return 0;
}
```

```
Enter academic marks: 98
Enter sports score: 78

Academic Marks: 98
Sports Score: 78
Total: 176

--------------------------------
Process exited after 12.01 seconds with return value 0
Press any key to continue . . .
```

Q4. Create a C++ program to demonstrate Hierarchical Inheritance.
Create a base class Vehicle containing:
Brand ,Model
Create two derived classes:
Car → Number of Doors
Bike → Engine Capacity
Create separate objects of Car and Bike and display their respective details.

```
#include <iostream>
using namespace std;
class Vehicle
{
protected:
    string brand;
    string model;
```

```
public:
    void acceptVehicle()
    {
        cout << "Enter brand: ";
        getline(cin, brand);

        cout << "Enter model: ";
        getline(cin, model);
    }
};

class Car : public Vehicle
{
private:
    int doors;

public:
    void acceptCar()
    {
        acceptVehicle();

        cout << "Enter number of doors: ";
        cin >> doors;
    }

    void displayCar()
    {
        cout << "\nCar Details:" << endl;
        cout << "Brand: " << brand << endl;
        cout << "Model: " << model << endl;
        cout << "Number of doors: " << doors << endl;
    }
};

class Bike : public Vehicle
{
private:
    int engineCapacity;

public:
    void acceptBike()
    {
        cin.ignore();

        acceptVehicle();

        cout << "Enter engine capacity: ";
        cin >> engineCapacity;
    }

    void displayBike()
    {
        cout << "\nBike Details:" << endl;
```

```
        cout << "Brand: " << brand << endl;
        cout << "Model: " << model << endl;
        cout << "Engine Capacity: " << engineCapacity << " cc" << endl;
    }
};

int main()
{
    Car c;
    Bike b;

    cout << "Enter Car Details" << endl;
    c.acceptCar();

    cout << "\nEnter Bike Details" << endl;
    b.acceptBike();

    c.displayCar();
    b.displayBike();

    return 0;
}
```

```
Enter Car Details
Enter brand: Toyota
Enter model: fortuner
Enter number of doors: 4

Enter Bike Details
Enter brand: honda
Enter model: shine
Enter engine capacity: 125

Car Details:
Brand: Toyota
Model: fortuner
Number of doors: 4

Bike Details:
Brand: honda
Model: shine
Engine Capacity: 125 cc

--------------------------------
Process exited after 73.04 seconds with return value 0
Press any key to continue . . .
```

Q5. Create a base class College with a constructor that initializes the College Name. Create a derived class Department with a constructor that initializes the Department Name. Create an object of the derived class and display both college and department information to show the order of constructor execution.

```
#include <iostream>
using namespace std;
```

```
class College
{
protected:
    string collegeName;

public:
    College(string c)
    {
        collegeName = c;
        cout << "College constructor called" << endl;
    }
};

class Department : public College
{
private:
    string departmentName;

public:
    Department(string c, string d) : College(c)
    {
        departmentName = d;
        cout << "Department constructor called" << endl;
    }

    void display()
    {
        cout << "\nCollege: " << collegeName << endl;
        cout << "Department: " << departmentName << endl;
    }
};

int main()
{
    Department d("ABC College", "Computer Science");

    d.display();

    return 0;
}
```

```
College constructor called
Department constructor called

College: ABC College
Department: Computer Science

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Process exited after 10.38 seconds with return value 0
Press any key to continue . . .
```